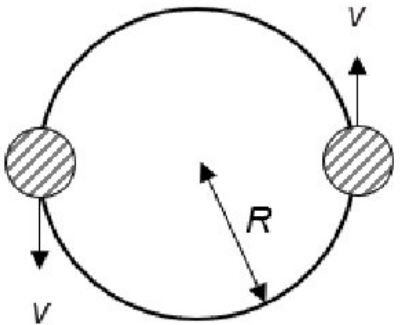
9 Two binary stars of equal mass M move with constant speed v in a circular orbit of radius R about their common centre of mass.



What is the speed v of each star?

A $\sqrt{\frac{GM}{4R}}$ **B** $\sqrt{\frac{GM}{2R}}$ **C** $\sqrt{\frac{GM}{R}}$ **D** $\sqrt{\frac{2GM}{R}}$